We start with the classical definition of parallel efficiency:

$$E=\frac{T_1}{T_1+T_o} \quad ,$$

where T_1 is the execution (processing) time of a volume of computational work spent by 1 CPU and measured in seconds, T_o – is the execution (processing) overhead time needed to complete this volume of work by p CPUs and measured in seconds.

By definition, this time scales are related to T_p (the execution (processing) time needed to complete this volume of work by p CPUs) via the relation:

$$pT_p = T_1 + T_o \quad .$$

And if we depict a volume of computational work as M (measured in bits) and processing throughput of a node as G (bits/s), then $T=M/G$. By supposing a situation, where the same amount of work M/p is distributed among different processing nodes which complete this work within interval T_k , we get:

$$pT_p = \sum_k T_k = \frac{M}{p} \sum G_{p,k}^{-1} \quad ,$$

and efficiency E can be calculated from the measurements of $G_{p,k}$ and G_1 as:

$$E = \frac{pG_1^{-1}}{\sum_k G_{p,k}^{-1}} \quad .$$

Fig. 1 presents the distribution of achieved throughput vs offered load (points 1-6 correspond to 1 containers pair / 1 CPU, points 7 – 13 to 2, 3, 4, 5, 6, 7, 8 forwarding processes). Fig. 2 presents the distribution of the mean latency vs offered load (points 1-6 correspond to 1 containers pair / 1 CPU, points 7 – 13 to 2, 3, 4, 5, 6, 7, 8 forwarding processes). Fig. 3 presents the distribution of the maximum latency vs offered load (points 1-6 correspond to 1 containers pair / 1 CPU, points 7 – 13 to 2, 3, 4, 5, 6, 7, 8 forwarding processes). Fig. 4 shows computational efficiency achieved on the system for these computational experiments (point 1 on the figure corresponds to the point 4 on Fig. 1, points 2-8 correspond to points 7-13 on Fig. 1).

The computational system involved in the experiments had: 2 AMD CPU cores with 4 logical cores (threads) totally, 12 Gb RAM, Ubuntu 22.04 with kernel 6.8. Three SDN data planes are compared: OVS, OpenSDN and Cilium.

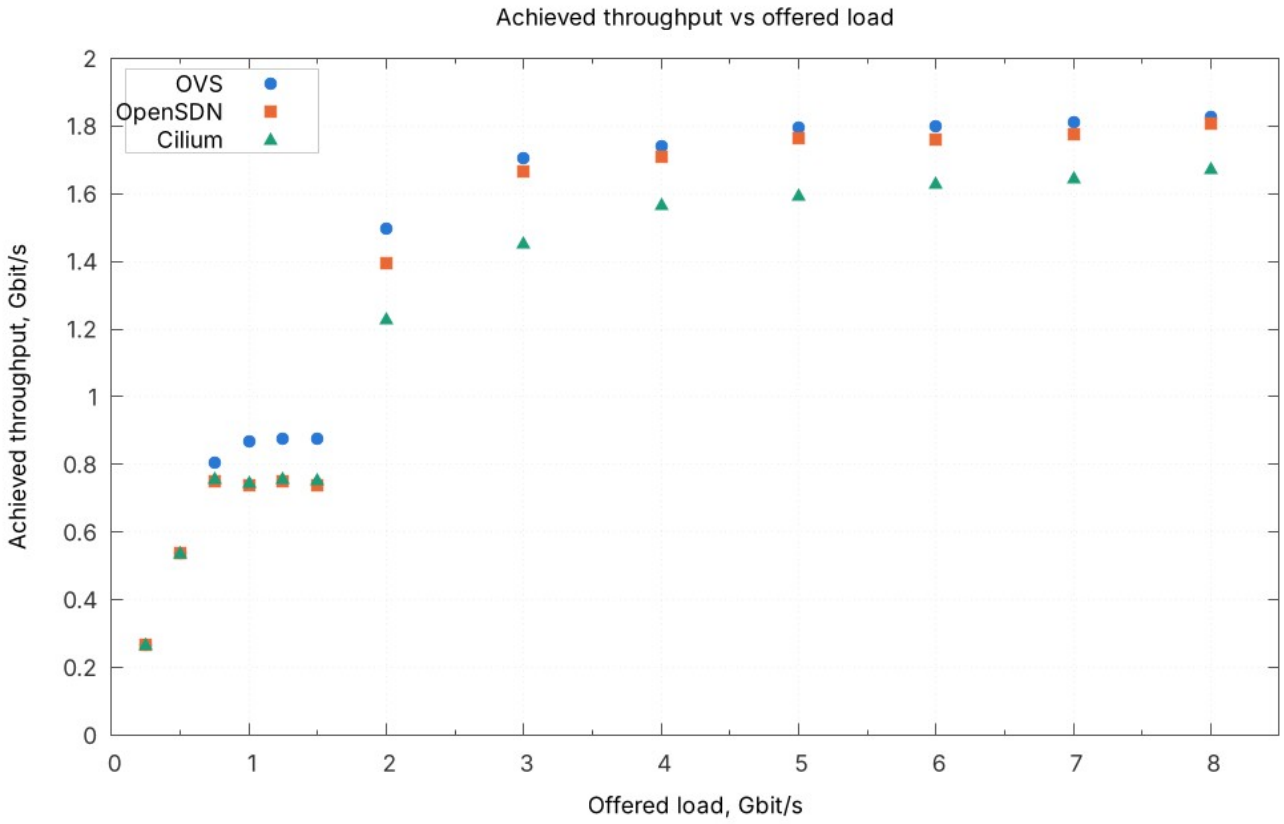

Achieved throughput vs offered load
OVS
OpenSDN
Cilium
Achieved throughput, Gbit/s
Offered load, Gbit/s
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
1
2
3
4
5
6
7
8

Fig. 1.

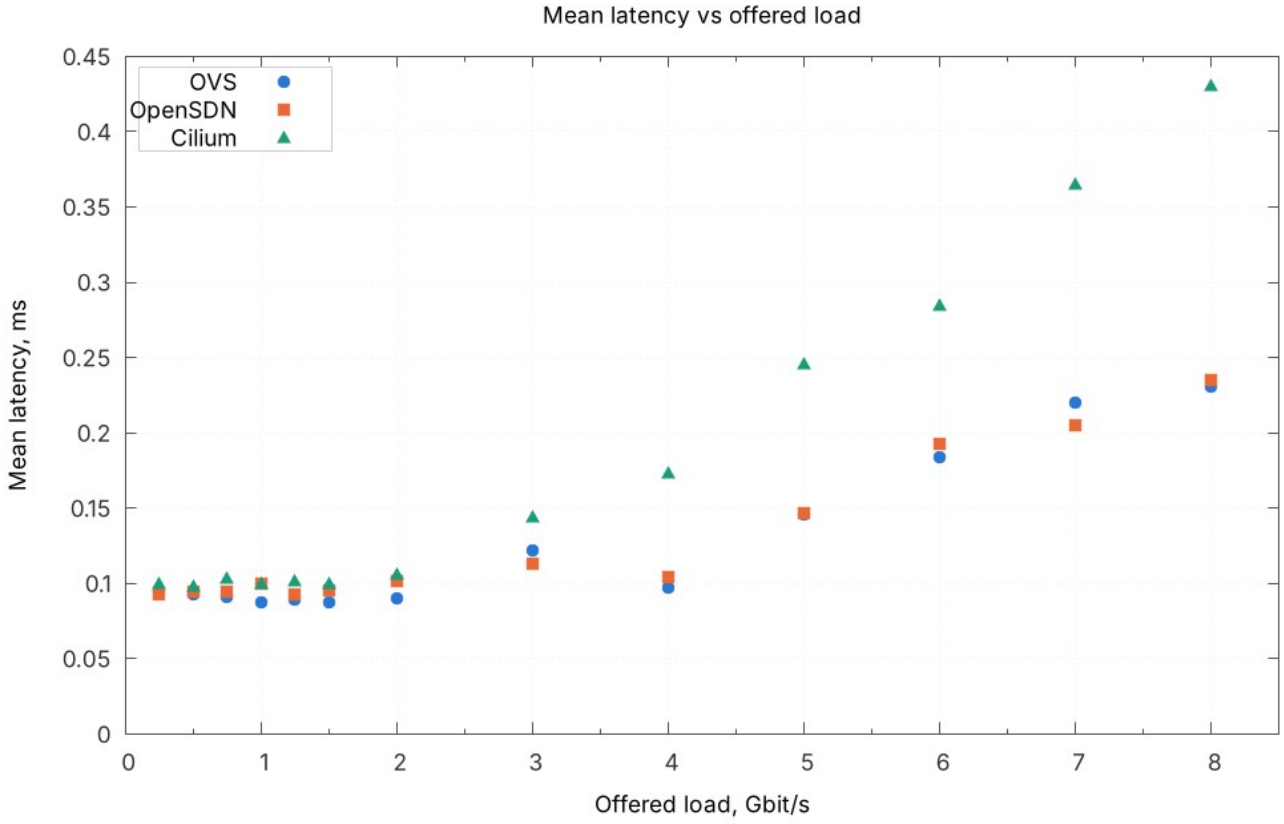

Mean latency vs offered load
OVS
OpenSDN
Cilium
Mean latency, ms
Offered load, Gbit/s
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0
1
2
3
4
5
6
7
8

Fig. 2.

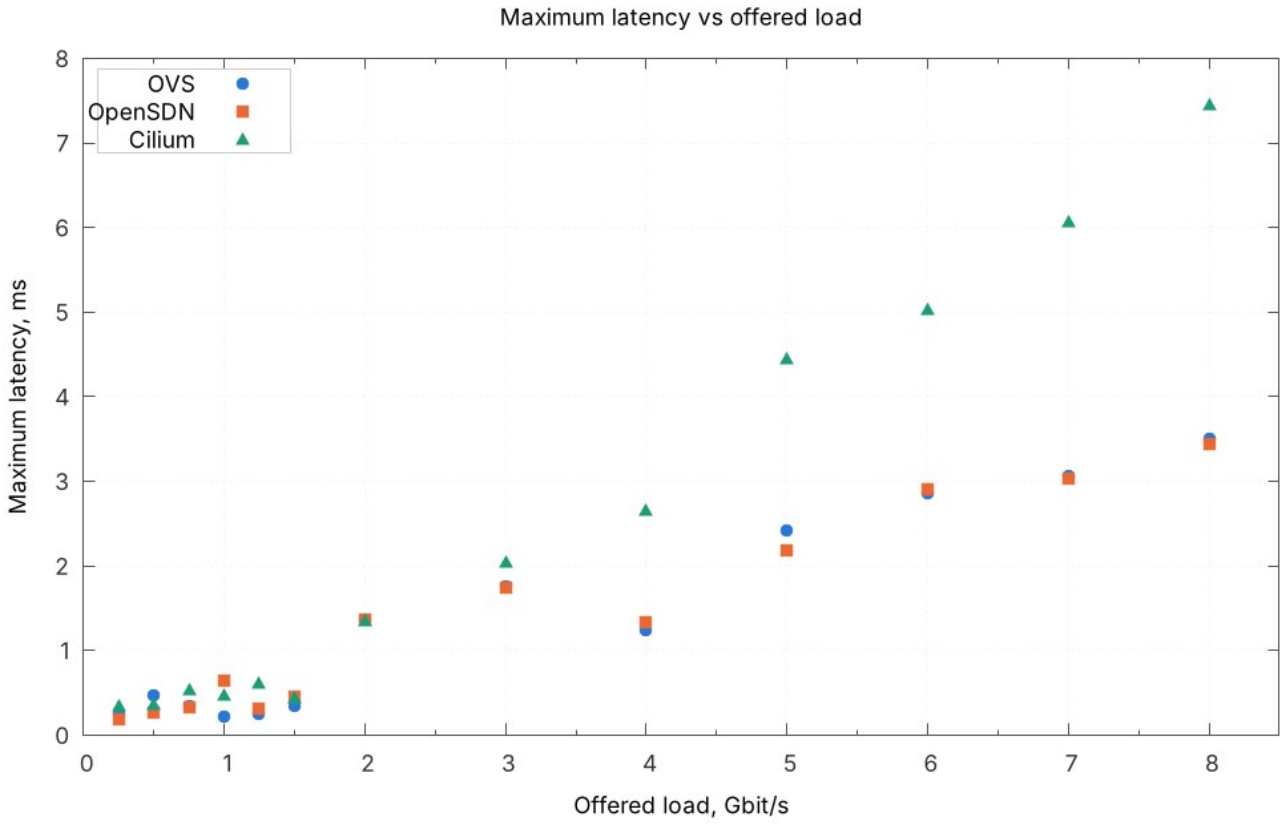

Maximum latency vs offered load
OVS
OpenSDN
Cilium
Maximum latency, ms
Offered load, Gbit/s
0
1
2
3
4
5
6
7
8

Fig. 3.

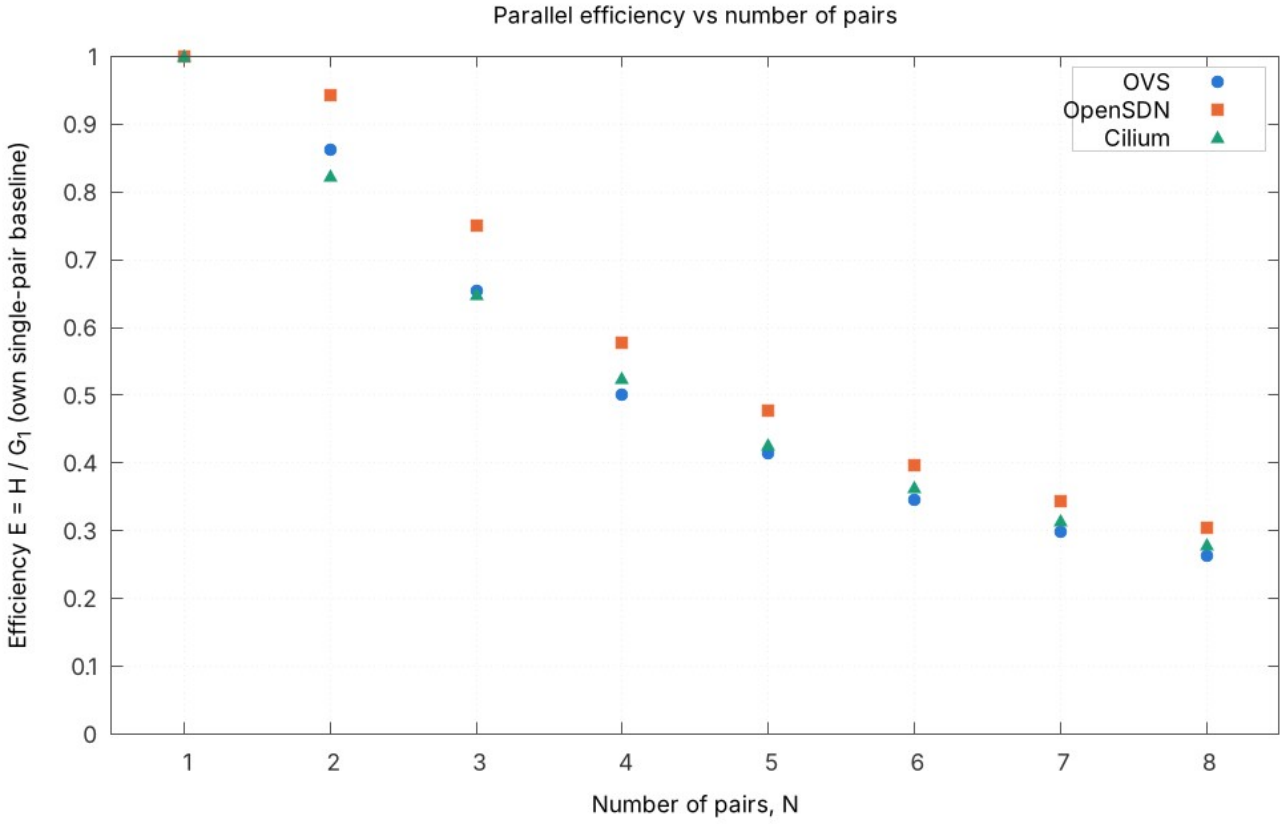

Parallel efficiency vs number of pairs
OVS
OpenSDN
Cilium
Efficiency E = H / G1 (own single-pair baseline)
Number of pairs, N
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
2
3
4
5
6
7
8

Fig. 4.